Attitude Estimation and Sensor Fusion Architecture for CAN-7USAT Flight Computer

The accurate determination of spatial orientation in a sub-orbital CanSat profile—characterized by violent ejection shocks exceeding 30 g, sustained spin stabilization, and rapid aerodynamic descent—demands rigorous sensor mechanization. An analysis of the specified flight computer architecture, leveraging a 600 MHz Cortex-M7 processor and an InvenSense ICM-20948 Inertial Measurement Unit (IMU), reveals several critical mathematical and configuration anomalies.

The primary symptoms observed in the ground station rendering—specifically, the left/right mirrored spatial rendering and the severe instability at high pitch angles (84.4°)—are classical indicators of reference frame reflections, omitted die-level axis transformations, and the mathematical singularities inherent to Euler angle parameterization. The following analysis systematically deconstructs the hardware configuration, frame transformations, algorithmic topologies, and calibration procedures required to achieve aerospace-grade attitude estimation, explicitly addressing the failures in the current configuration.

1. True Sensor Axis Convention and Specific Force Polarity

Establishing a rigorously defined coordinate frame is the foundational step for any Strapdown Inertial Navigation System (SINS). Misinterpreting the manufacturer's axis definitions virtually guarantees downstream failure in sensor fusion algorithms.

According to the primary TDK InvenSense ICM-20948 datasheet (DS-000189), the orientation of the accelerometer and gyroscope axes of sensitivity is strictly defined relative to the package's physical Pin 1 identifier[1]. Observing the package directly from above with the Pin 1 dot positioned in the top-left corner:

- The **+X axis** points directly to the right.
- The **+Y axis** points directly up (towards the Pin 1 edge).
- The **+Z axis** points orthogonally out of the package (upwards, away from the printed circuit board).

This defines a standard, right-handed Cartesian coordinate system. Applying the right-hand rule, the cross product of the +X vector (right) and the +Y vector (up) yields a +Z vector pointing out of the plane towards the observer[1]. The 7Semi breakout board's silkscreen should reflect this exact alignment, though silkscreens frequently contain errors; the physical package dot remains the absolute ground truth.

Specific Force and Accelerometer Sign Convention

Microelectromechanical systems (MEMS) accelerometers do not directly measure kinematic acceleration; they measure specific force, which is the non-gravitational force acting on the sensor's proof mass[2]. When the device is held perfectly stationary on a bench, the true kinematic acceleration is zero. The sensor measures the normal force exerted by the bench to counteract gravity and keep the proof mass from falling toward the Earth's center of mass. Therefore, when a specific sensor axis points **UP** (away from the Earth's center of mass), the sensor must register a positive value. Specifically, for the $\pm 16g$ full-scale range (FSR) where the sensitivity is 2048 LSB/g:

- **+Z Pointing Up (Away from Earth):** Reads $+1g$ (approximately $+2048$ LSB).
- **+Z Pointing Down (Toward Earth):** Reads $-1g$ (approximately -2048 LSB).
- **+X Pointing Up:** Reads $+1g$.
- **+X Pointing Down:** Reads $-1g$.

This exact polarity must be mathematically enforced in the flight software. The current configuration utilizes a scale factor of 32768 / 16 LSB per g, generating normalized outputs. If the sign convention is inverted in the firmware, the gravity vector will be projected incorrectly into the body frame, forcing the attitude estimator to converge on an inverted mathematical hemisphere, producing inverted pitch or roll tracking.

2. Magnetometer Axis Alignment Trap

A critical architectural detail of the ICM-20948 is that it is a Multi-Chip Module (MCM). The 24-pin QFN package houses two physically separate silicon dies: the primary InvenSense accelerometer/gyroscope die and the Asahi Kasei Microdevices (AKM) AK09916 magnetometer die[1]. Because these dies are independently bonded to the same substrate, their internal coordinate frames are not natively aligned.

As documented in the orientation diagrams of the ICM-20948 datasheet (Figure 12 for the IMU and Figure 13 for the Magnetometer), the AK09916 frame differs fundamentally from the primary IMU frame[1]. Relative to the same Pin 1 top-left reference:

- The Magnetometer **+X axis** points to the right (aligned with Accel/Gyro +X).
- The Magnetometer **+Y axis** points down (inverted relative to Accel/Gyro +Y).
- The Magnetometer **+Z axis** points into the package (inverted relative to Accel/Gyro +Z).

While legacy devices such as the MPU-9250 required a swap of the X and Y axes[4], the AK09916 within the ICM-20948 requires a strict negation of the Y and Z axes to map its readings into the primary IMU frame. The exact mathematical transformation required to align the raw magnetometer readings (m_{raw}) to the accelerometer/gyroscope frame ($m_{aligned}$) is:

$$m_{aligned,x} = m_{raw,x} \quad m_{aligned,y} = -m_{raw,y} \quad m_{aligned,z} = -m_{raw,z}$$

If this transformation is omitted in the raw register polling architecture, the Y and Z magnetic axes remain inverted relative to the inertial sensors. Because flipping exactly two axes

preserves the right-handedness of the coordinate system, this does not create a left-handed frame in isolation. However, the resulting vectors are rotated 180° around the X-axis. When an unaligned magnetometer vector is fed into a tilt-compensated heading calculation (using roll and pitch derived from the correctly oriented accelerometer), the horizontal components of the magnetic field are projected incorrectly. The inversion of the Y-axis causes the computed yaw to mirror; a physical clockwise rotation of the vehicle will result in an anti-clockwise rotation of the computed yaw angle. This AK09916 axis misalignment trap is a documented and frequent cause of mirrored left/right rendering in 3D ground station visualizations when implementing custom drivers[5].

3. Estimator Degeneracy and Filter Selection

The observation of severe instability at specific orientations (pitch readings of 84.4° and tilt of 88.2°), combined with left/right mirrored output, points directly to a mathematical degeneracy in the attitude representation. The current architecture utilizes a complementary filter formulated with Tait-Bryan Euler angles.

The Gimbal Lock Singularity

A three-dimensional rotation can be parameterized as a sequence of three elemental rotations. In aeronautics, the standard Z-Y-X sequence applies yaw (ψ) about the Z-axis, pitch (θ) about the intermediate Y-axis, and roll (ϕ) about the resulting X-axis.

The kinematic equation mapping measured body rates ($\omega_x, \omega_y, \omega_z$) to Euler angle rates ($\dot{\phi}, \dot{\theta}, \dot{\psi}$) requires inverting the Euler transformation matrix:

$$\begin{bmatrix} \dot{\phi} \\ \dot{\theta} \\ \dot{\psi} \end{bmatrix} = \begin{bmatrix} 1 & \sin\phi \tan\theta & \cos\phi \tan\theta \\ 0 & \cos\phi & -\sin\phi \\ 0 & \sin\phi \sec\theta & \cos\phi \sec\theta \end{bmatrix} \begin{bmatrix} \omega_x \\ \omega_y \\ \omega_z \end{bmatrix}$$

As the vehicle's pitch (θ) approaches $\pm 90^\circ$ (a perfectly vertical orientation), the terms $\tan\theta$ and $\sec\theta$ approach infinity. At exactly 90°, the roll and yaw axes become perfectly collinear. The system loses a degree of freedom, and any perturbation in the X-Z plane cannot be uniquely resolved into roll or yaw[5]. This mathematical singularity, known as gimbal lock, forces the equations to violently swap roll and yaw values, resulting in sudden 180° sign flips. This produces the exact mirrored behavior observed near the vertical axis. Because a CanSat descends vertically and typically tumbles through zenith and nadir during deployment, Euler angles are fundamentally disqualifying for this mission profile.

Attitude Estimator Topology Comparison

To survive tumbling and vertical orientations, the attitude must be parameterized using quaternions (q_0, q_1, q_2, q_3), which provide a globally non-singular representation of SO(3) rotational space[8]. The optimal filter topology depends heavily on the computational capabilities of the microcontroller.

Filter Topology	Computational Cost (Cortex-M7)	Convergence / Tuning	Shock / Spin Behavior	Bias Tracking
Euler Complementary	Minimal (< 2 μ s)	Single gain parameter	Fails at $\pm 90^\circ$ pitch. Fails under 30 g shock without gating.	None. Integrates drift indefinitely.
Madgwick (Quaternion)	Low (~10 μs)	Step size parameter (β)	Robust through singularities. Susceptible to acceleration shock corruption if β is fixed.	Yes, coupled to gradient descent[9].
Mahony (Quaternion)	Low (~8 μs)	Proportional (K_p), Integral (K_i)	Highly robust. Explicit high-shock rejection is easily achieved via conditional gain scaling.	Explicit independent integration of gyro bias[10].
Extended Kalman Filter	Medium (~50 μs)	Process (Q) and Measurement (R) noise	Optimal. Dynamically scales measurement covariance R	Excellent, tracks 3×3 covariance of bias states.

		covariance matrices	during 30 g shock to ignore accelerometer.	

The Teensy 4.1's 600 MHz Cortex-M7 utilizes an NXP i.MX RT1062 processor featuring a hardware Floating Point Unit (FPU) capable of single-cycle 32-bit floating-point execution and dual-issue superscalar instruction pipelining[11]. Given this vast computational headroom, a 7-state Extended Kalman Filter (EKF)—comprising four quaternion states and three gyroscope bias states—is the theoretical optimum[9].

However, implementing an EKF from scratch introduces severe complexity involving Jacobian matrix derivations and matrix inversions. The practical recommendation for a CanSat is the **Explicit Complementary Filter on the Special Orthogonal Group SO(3)**, universally known as the **Mahony Filter**[13]. It matches the EKF in steady-state accuracy but requires no matrix inversions.

The Mahony algorithm utilizes a proportional-integral (PI) controller framework. The cross-product of the estimated gravity vector (derived from the current quaternion) and the measured gravity vector (from the accelerometer) drives an error term. The proportional gain (K_p) corrects immediate attitude, while the integral gain (K_i) explicitly estimates and continuously removes gyroscope bias during flight[10]. Crucially, the Mahony filter accommodates a dynamic gating mechanism for the 30 g ejection shock. By calculating the magnitude of the accelerometer vector $|a|$, the filter can set $K_p = 0$ and $K_i = 0$ whenever $||a| - 1.0| > 0.25g$, relying entirely on the spin-stabilized gyroscope data during the shock impulse without corrupting the quaternion state.

4. Empirical Determination of Mounting Orientation

Flipping individual signs iteratively is a divergent strategy. To definitively diagnose the mirrored left/right symptom, the exact rotation matrix mapping the IMU package frame to the CanSat vehicle body frame must be empirically measured and verified.

The Six-Position Static Test

The vehicle must be held perfectly static in six orthogonal orientations relative to gravity. Assume a standard aerospace vehicle body frame: +X (Nose/Forward), +Y (Right), +Z (Down).

At each position, record the averaged accelerometer vector $[a_x, a_y, a_z]^T$ from the sensor over a 3-second window, normalized to $1g$.

Position	**Physical Vehicle Orientation**	**Expected Body Frame Gravity Vector Vbody**

1	Nose Down (Vertical)	$[+1, 0, 0]^T$
2	Nose Up (Vertical)	$[-1, 0, 0]^T$
3	Right Side Down	$[0, +1, 0]^T$
4	Left Side Down	$[0, -1, 0]^T$
5	Bottom Down (Normal stance)	$[0, 0, +1]^T$
6	Top Down (Inverted)	$[0, 0, -1]^T$

To derive the 3×3 direction cosine matrix (DCM) R_{mount} that maps the raw sensor readings to the body frame, select three linearly independent measurements (e.g., Positions 1, 3, and 5). Construct a 3×3 matrix M_{sensor} where each column is the normalized sensor vector recorded for that position. Construct a corresponding M_{body} matrix using the expected vectors from the table above. The rotation matrix is derived via:

$$R_{mount} = M_{body} \cdot M_{sensor}^{-1}$$

Validation and the Determinant Trap

For R_{mount} to represent a valid, rigid-body rotation in 3D space, it must be strictly orthonormal ($R^T R = I$) and its determinant must exactly equal $+1$. Calculate the determinant:

$$\det(R) = R_{11}(R_{22}R_{33} - R_{23}R_{32}) - R_{12}(R_{21}R_{33} - R_{23}R_{31}) + R_{13}(R_{21}R_{32} - R_{22}R_{31})$$

.

If $\det(R) \approx -1$, the calculated transformation represents a mathematical **reflection**, meaning a left-handed coordinate frame has been accidentally synthesized[2]. A determinant of -1 is the most probable systemic cause of the current bug. Feeding a left-handed frame into a 3D ground station renderer expecting a right-handed system will yield exactly the mirrored left/right movement described. Resolving this requires ensuring the body axes definition rigorously follows the right-hand rule and that the mapping matrix enforces $\det(R) = 1$.

For the gyroscope, empirical validation involves recording the integrated angular rate while manually rotating the vehicle exactly $+90^\circ$ about each defined body axis. The raw integrated body rates multiplied by R_{mount} must yield exactly $+90^\circ$ on the targeted body axis and near zero on the orthogonal axes.

5. Comprehensive Sensor Calibration Plan

An uncalibrated MEMS sensor guarantees attitude divergence. The calibration sequence must isolate deterministic errors (biases, scale factors, non-orthogonality) from stochastic errors (noise, random walks).

Gyroscope Calibration

- **Static Bias Estimation:** Gyroscope bias is heavily temperature-dependent[16]. On the bench, power the Teensy and ICM-20948 and allow it to thermally stabilize for at least 5 minutes. Average the raw gyroscope readings in all three axes for 60 seconds while completely undisturbed to yield the static zero-rate offset.
- **In-Flight Re-estimation:** Because internal CanSat temperatures fluctuate dynamically during ascent and deployment, relying solely on a bench bias will lead to drift. The Mahony filter's integral gain (K_i) must be tuned to continually track and update this bias during the parachute descent[10].

Accelerometer Calibration

- **Six-Position Method:** The factory trim is insufficient for precise tilt estimation. The previously described six-position test must be expanded into a Linear Least Squares solver to determine both the 3-axis biases and the 3-axis scale factors[2]. Because the mission profile requires gravity vector finding with an accuracy target of $\approx 1^\circ$, estimating complex cross-axis misalignment matrices is unnecessary; a 6-parameter offset/scale fit is sufficient.

Magnetometer Hard and Soft Iron Calibration

The presence of the brushed DC N20 motor, structural aluminum, and high-current battery traces natively warps the local magnetic field.

- **Procedure:** The magnetometer must be calibrated *after* it is permanently mounted inside the fully assembled CanSat[18].
- **Execution:** Log raw magnetometer data while tumbling the assembled vehicle in 3D space, drawing a complete sphere of data points (minimum 1,000 samples capturing all orientations).
- **Ellipsoid Fitting:** Uncalibrated magnetic data forms an offset ellipsoid. Hard-iron errors (permanent local magnetic fields) shift the center of the ellipsoid away from the origin; soft-iron errors (induced magnetism in ferrous metals) distort it from a sphere into an ellipsoid[20]. Using a nonlinear least-squares regression algorithm (e.g.,

Levenberg-Marquardt), fit the data to the equation of an ellipsoid to extract the translation vector (hard iron offset) and the 3×3 transformation matrix (soft iron scale/misalignment) required to warp the data back into a perfect sphere centered at $(0, 0, 0)$ [19].

Allan Variance (IEEE Std 952)

Allan variance is a time-domain analysis technique used to characterize the stochastic noise properties of inertial sensors by evaluating the root-mean-square random drift error as a function of averaging time τ [21]. To compute it, log high-rate static gyroscope data for at least 2 hours. Calculate the variance of the difference between successive data clusters of length τ, and plot the results on a log-log scale.

- **Angle Random Walk (ARW):** Found where the slope of the curve is -0.5 (typically at $\tau = 1$ s). It characterizes high-frequency white noise[23].
- **Bias Instability (BI):** Found at the lowest, flat point of the curve (slope 0). This is the absolute theoretical limit of the sensor's accuracy, representing low-frequency flicker noise[24].

For a 20-minute CanSat flight, integrating the ARW dictates the immediate random drift, while the BI dictates the slowly wandering zero-point. The maximum expected yaw drift can be predicted using $Drift_{yaw} = ARW \cdot \sqrt{T} + BI \cdot T$. If the Allan Variance reveals a Bias Instability of 10°/hour, the maximum uncorrected yaw drift over a 20-minute flight will remain safely under a few degrees, mathematically validating the hardware's suitability for the mission profile.

6. Digital Motion Processor (DMP) Feasibility

The ICM-20948 features an on-chip Digital Motion Processor designed to offload sensor fusion, capable of outputting 6-axis or 9-axis quaternions at up to 225 Hz[1]. However, for a high-performance aerospace application built on a Teensy 4.1, relying on the DMP is overwhelmingly disadvantageous.

1. **Firmware Payload & Boot Delay:** The DMP is not hardcoded in silicon; it requires the host microcontroller to push a 14 kilobyte proprietary firmware blob across the I²C bus during every boot cycle[26]. This significantly delays system startup and consumes flash memory.
2. **Algorithmic Opacity:** The sensor fusion algorithms inside the DMP are undocumented "black boxes"[26]. It is impossible to explicitly adjust the algorithm's shock rejection thresholds, dynamically alter filter gains during the 30 g ejection, or monitor the internal covariance matrices.
3. **Reliability Issues:** Embedded engineering communities repeatedly document severe

bugs with the ICM-20948 DMP, notably sudden yaw drifts, magnetic heading corruption when pitched beyond 45°, and spontaneous coordinate frame shifts[27].

The Cortex-M7 executes an optimized C++ Mahony filter in under 10μ s. Relinquishing deterministic control of the state estimator to an undocumented, closed-source coprocessor is an unnecessary engineering risk that limits troubleshooting capabilities and removes the ability to defend the mathematical models to competition judges. The DMP should remain disabled.

7. Configuration and Tuning Sanity Check

Evaluating the specific parameters provided in the current configuration reveals several compromises that must be rectified.

Complementary Filter Time Constant

A complementary filter coefficient of $\alpha = 0.98$ running at an update rate of 50 Hz ($dt = 0.02$ s) yields a specific time constant τ. The continuous-time derivation relates these variables via:

$$\alpha = \frac{\tau}{\tau + dt} \implies 0.98 = \frac{\tau}{\tau + 0.02} \implies 0.98\tau + 0.0196 = \tau \implies 0.02\tau = 0.0196 \implies \tau = 0.98 \text{ seconds}$$

[29]. A time constant of ~1 second is appropriate for steady-state descent under a parachute, as it suppresses short-term translational accelerations while smoothly tracking the long-term gravity vector. However, a static complementary filter fails critically during rapid spin stabilization, whereas a Mahony filter's dynamically adjusting proportional gain handles sustained spin seamlessly.

Digital Low Pass Filter (DLPF) and Aliasing

The gyroscope is configured with GYRO_DLPFCFG = 4, yielding a hardware low-pass cutoff frequency of 23.9 Hz[1]. The data is polled at a 50 Hz service tick. According to the Nyquist-Shannon sampling theorem, the Nyquist frequency is exactly 25 Hz. A 23.9 Hz low-pass filter provides virtually zero attenuation at the 25 Hz boundary. Any motor vibrations, gear noise from the N20 motor, or aero-acoustic noise above 25 Hz will fold back into the baseband, permanently corrupting the rate data through aliasing. **Correction:** Either increase the polling rate to a minimum of 100 Hz, or lower the DLPF cutoff to 11.9 Hz (GYRO_DLPFCFG = 5) to provide an adequate anti-aliasing guard band[30].

Full-Scale Ranges and Quantization Noise

- **Gyroscope (± 2000 dps):** A ± 2000 dps range mapped across a 16-bit ADC yields a resolution of exactly 16.4 LSB/dps[31]. If the vehicle spins at 1500 dps during stabilization, this FSR is mandatory. However, if the sustained spin is only 300 dps, operating at ± 2000 dps artificially discards two bits of ADC resolution, worsening the quantization noise. Calculate the exact anticipated spin rate based on the motor RPM and wheel inertia, and drop the FSR as low as mathematically possible without clipping.

- **Accelerometer ($\pm 16g$):** An ejection shock of 30 g will instantly saturate the sensor at ± 32767 LSB. If an attitude filter ingests a clipped vector, the gravity estimate diverges immediately. The current shock-rejection gate (ignoring $|a|$ outside $0.75 - 1.25g$) is the mathematically correct mechanism to survive this saturation[32]. The $\pm 16g$ limit is unavoidable for the ICM-20948, but the software mitigation is sound.

I²C Bus Speed and Clock Stretching Lockups

The report notes that testing at 400 kHz "broke the part." The Teensy 4.1's hardware I²C implementation relies on exceptionally weak internal pull-up resistors (typically $> 30\text{k}\Omega$)[34]. At 400 kHz, bus capacitance combined with weak pull-ups causes critically slow rise times. This violates the I²C specification, leading the ICM-20948 to misinterpret clock edges, resulting in clock stretching failures and catastrophic bus lockups[35]. **Correction:** Install stiff external pull-up resistors (e.g., 2.2 kΩ) connected to 3.3V on both the SDA and SCL lines. This will guarantee sharp signal edges, safely enabling 400 kHz operation and reducing the time the CPU wastes waiting on the bus.

8. Verdict on the Mirrored Symptom

The mirrored left/right movement combined with an $80^\circ+$ singularity collapse is unequivocally caused by a combination of compounding errors. Ranked from most to least likely:

1. **Gimbal Lock (Euler Singularity):** The primary cause of the 84.4° collapse. As the pitch approaches 90°, the Euler matrix inversion drives roll and yaw to infinity, causing sudden 180° phase flips that render left and right inverted on the ground station.
2. **Determinant -1 (Left-Handed Frame):** The primary cause of persistent mirrored tracking. Manually flipping individual axis signs without ensuring the rotation matrix determinant equals $+1$ synthesizes a left-handed coordinate frame, which mirrors all rotations when fed into a standard right-handed 3D renderer.
3. **Magnetometer Misalignment:** If the magnetometer is eventually fused, omitting the Y and Z inversions required by the AK09916 die will project the magnetic vector backwards across the X-axis, mirroring the yaw calculation.

The single measurement that discriminates between these is the rotation matrix determinant. If $\det(R_{mount}) = -1$, the frame is fundamentally broken. If $\det(R_{mount}) = 1$, the mirrored output is solely a byproduct of the Euler gimbal lock.

9. Explicit List of Current Configuration Failures

Referencing the provided configuration, the following elements are fundamentally incorrect and must be addressed:

1. **Attitude Filter Topology:** The atan2 complementary filter suffers from Euler singularities and lacks explicit gyro bias tracking. It must be replaced with a quaternion-based Mahony filter.
2. **DLPF Cutoff Frequency:** GYRO_DLPFCFG = 4 (23.9 Hz) at a 50 Hz poll rate violates Nyquist margins, guaranteeing vibration aliasing.
3. **Magnetometer Transform:** The AK09916 bypass code polls 9 bytes but fails to invert the Y and Z axes to match the IMU frame.
4. **I²C Hardware Design:** Operating at 100 kHz wastes CPU cycles. The 400 kHz failure indicates missing or insufficient external pull-up resistors on the I²C bus.
5. **Heuristic Sign Flipping:** Guessing axis signs synthesizes invalid left-handed matrices. The mounting orientation must be strictly derived via the 6-position deterministic method.

By migrating to quaternions, mathematically enforcing a right-handed reference frame via matrix determinants, adjusting the low-pass filters for anti-aliasing, and rigorously calibrating the hard-iron magnetic distortions in situ, the flight computer will achieve robust, continuous attitude tracking through high-g ejection, sustained spin, and vertical parachute descent.

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36. [solved] Teensy 4.1 call setClock AFTER mcp.begin to increase SCL-Cock-frequency, https://forum.arduino.cc/t/solved-teensy-4-1-call-setclock-after-mcp-begin-to-increase-scl-cock-frequency/876464